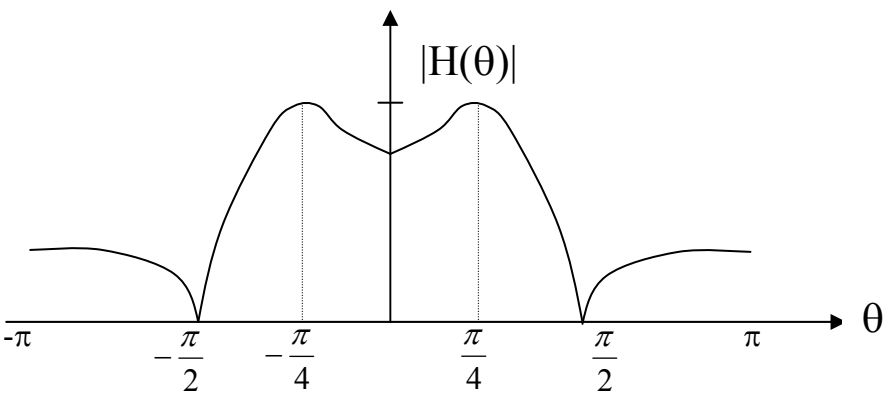
Sample Final Exam Paper
Model Answers

Question 1

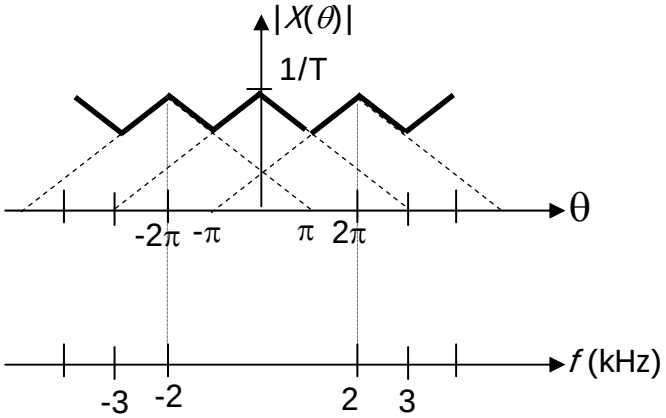
(a) (i)

$$H(z) = K\,\frac{(z+j)(z-j)}{(z-0.5-0.5j)(z-0.5+0.5j)} = K\,\frac{(1+z^{-2})}{1-z^{-1}+0.5z^{-2}}$$

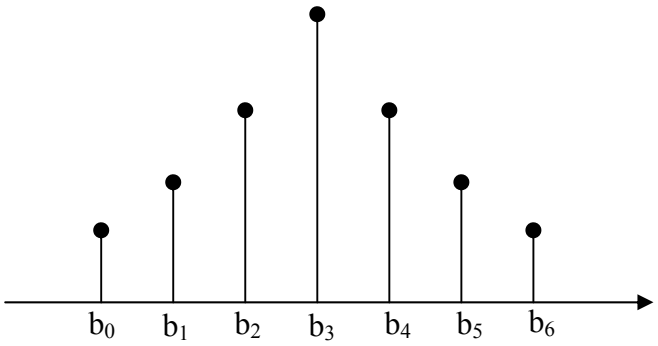
(ii)



(b)



(c) (i)



For a linear phase the coefficients must be symmetrical about b_3.
Therefore $b_4 = b_2 = 3$, $b_5 = b_1 = 2$, $b_6 = b_0 = 1$

(ii)

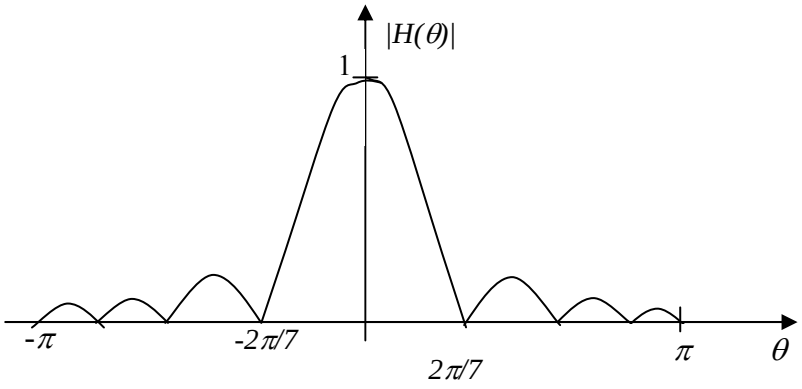
$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}$$

$$= \frac{1 - z^{-N}}{1 - z^{-1}} = \frac{1 - z^{-7}}{1 - z^{-1}} \qquad N = 7$$

$$H(\theta) = \frac{1 - e^{-j7\theta}}{1 - e^{-j\theta}} = \frac{e^{\frac{-j7\theta}{2}}}{e^{\frac{-j\theta}{2}}} \left[\frac{e^{\frac{j7\theta}{2}} - e^{\frac{-j\theta}{2}}}{e^{\frac{j\theta}{2}} - e^{\frac{-j\theta}{2}}} \right]$$

$$H(\theta) = e^{-j3\theta} \frac{\sin\frac{7\theta}{2}}{\sin\frac{\theta}{2}} \qquad \therefore |H(\theta)| = \left| \frac{\sin\frac{7\theta}{2}}{\sin\frac{\theta}{2}} \right|$$

$$\therefore \phi(0) = -3\theta$$



(iii)

$$\phi(0) = -3\theta \qquad \therefore \frac{-d\phi(\theta)}{d\theta} = 3 \leftarrow \text{Constant}$$

group delay

(d) (i) $H(z)$ is an IIR filter.

(ii)

$$H(z) = \frac{\frac{1}{2}(1 - az^{-1}) + 1}{(1 - \frac{1}{2}z^{-1})(1 - az^{-1})} = \frac{\frac{1}{2} - \frac{a}{2}z^{-1} + 1}{1 - az^{-1} - \frac{1}{2}z^{-1} + \frac{a}{2}z^{-2}}$$

$$= \frac{\frac{3}{2} - \frac{a}{2}z^{-1}}{\frac{2 - (2a+1)z^{-1} + az^{-2}}{2}} = \frac{3 - az^{-1}}{2 - (2a+1)z^{-1} + az^{-2}}$$

$$2y(n) - (2a+1)y(n-1) + ay(n-2) = 3x(n) - ax(n-1)$$

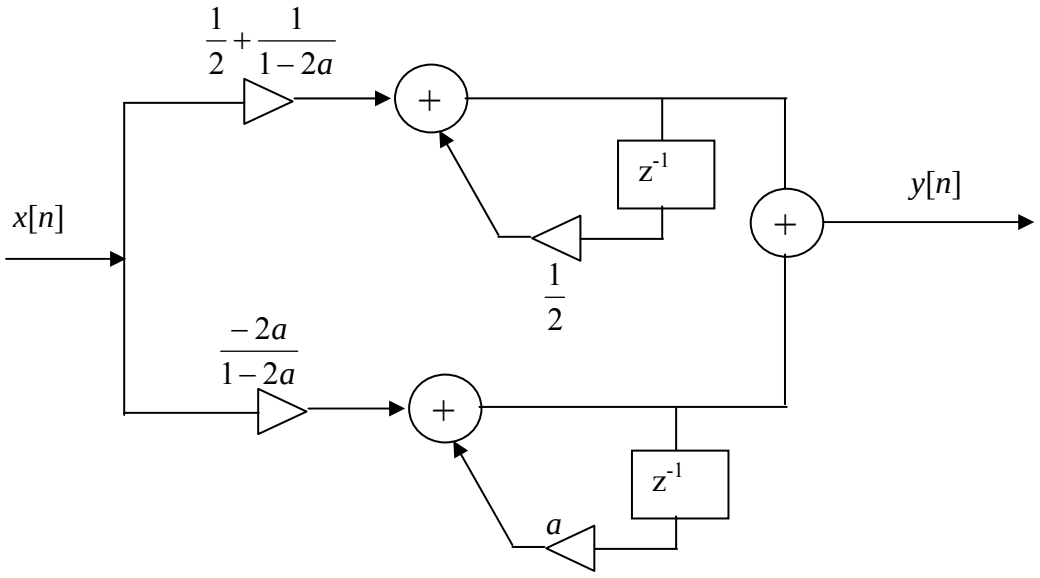
$$\therefore y(n) = \frac{3}{2}x(n) - \frac{a}{2}x(n-1) + \frac{2a+1}{2}y(n-1) - \frac{a}{2}y(n-2)$$

(iii)

$$H(z) = \frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - az^{-1}}$$

$$\frac{1}{(1 - \frac{1}{2}z^{-1})(1 - az^{-1})} = \frac{\frac{1}{1-2a}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{-2a}{1-2a}}{1 - az^{-1}} = \frac{\frac{1}{1-2a}}{1 - \frac{1}{2}z^{-1}} - \frac{\frac{2a}{1-2a}}{1 - az^{-1}}$$

$$H(z) = \frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{1-2a}}{1 - \frac{1}{2}z^{-1}} - \frac{\frac{2a}{1-2a}}{1 - az^{-1}} = \frac{(\frac{1}{2} + \frac{1}{1-2a})}{1 - \frac{1}{2}z^{-1}} + \frac{-(\frac{2a}{1-2a})}{1 - az^{-1}}$$



(iv)

$$h(n) = \left(\frac{1}{2} + \frac{1}{1-2a}\right)\left(\frac{1}{2}\right)^n u(n) - \left(\frac{2a}{1-2a}\right)a^n u(n)$$

When $a < 1$, $h(n)$ will decrease as n increases (stable filter).

If $a > 1$, $h(n)$ will increase as n increases (unstable filter).

Question 2:

(a) (i)

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b}{1 - az^{-1}}$$

$$\left|H(\theta)\right|_{\theta=0} = 1 \qquad \therefore 1 = \frac{b}{1-a}$$

$$\therefore b = 1 - a$$

$$\text{Normalised filter: } H(z) = \frac{1-a}{1-az^{-1}}$$

(ii)

$$\left|H(\theta)\right|^2 = \left|\frac{(1-a)^2}{(1-a\cos\theta)^2 + a^2\sin^2\theta}\right| = \frac{(1-a)}{1-a^2-2a\cos\theta}$$

Let the 3 dB cut-off be θc

$$\therefore \frac{1}{2}\left|H(0)\right|^2 = \left|H(\theta_C)\right|^2$$

$$\frac{1}{2}.1 = \frac{(1-a)^2}{1-a^2-2a\cos\theta_c} \quad \therefore 1 + a^2 - 2(1-a)^2 = 2a\cos\theta_c$$

$$\therefore \cos\theta_c = \frac{-a^2 + 4a - 1}{2a}$$

$$\theta_c = \cos^{-1}\left[\frac{-a^2 + 4a - 1}{2a}\right]$$

(b) (i) $H(z) = \dfrac{z}{z^2 - 2\cos\theta_0 z + 1} = \dfrac{z}{z^2 - b_1 z + 1}$

From z transform table: $\sin n\theta_0\ u(n) \leftrightarrow \dfrac{z\sin\theta_0}{z^2 - 2\cos\theta_0 z + 1}$

$$\therefore \frac{\sin(n\theta_0)}{\sin\theta_0} u(n) \leftrightarrow \frac{z}{z^2 - 2\cos\theta_0 z + 1}$$

H(z)

$$\therefore h(n) = \frac{\sin n_0}{\sin\theta} u(n)$$

(ii) $b_1 = 2\cos\theta_0$

therefore by differentiating with respect to theta we obtain

$\Delta b_1 = -2\sin\theta_0 . \Delta\theta_0$

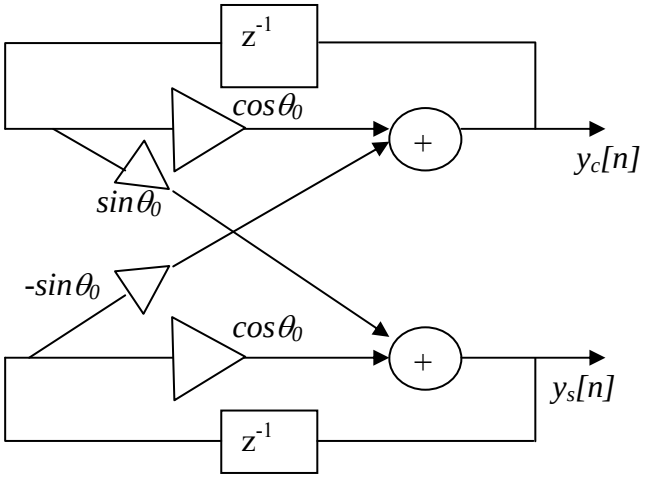
$\Delta b_1 = -2\sin\theta_0 . 2\pi \dfrac{\Delta f_0}{f_s}$ Therefore $\Delta f_0 = \dfrac{-f_s \Delta b_1}{4\pi \sin\left(\dfrac{2\pi f_0}{f_s}\right)}$

(iii) Δf_0 minimum, provides the high resolution when $\sin(2\pi \dfrac{f_0}{f_s}) = 1$

$$\therefore |\Delta f_0| = \frac{f_s \Delta b_1}{4\pi}$$

(c) (i) $y_c(n) = \cos\theta_0\ y_c(n-1) - \sin\theta_0\ y_s(n-1)$

$y_s(n) = \sin\theta_0\ y_c(n-1) + \cos\theta_0\ y_s(n-1)$



n = 0

(ii)

n = 0 $\quad y_s(0) = \sin\theta_0\,(A\cos\theta_0) + \cos\theta_0(A\sin\theta_0) = 0$

$y_c(0) = \cos\theta_0\,(A\cos\theta_0) - \sin\theta_0(-A\sin\theta_0) = A$

$y_c(0) = A$

n = 1 $\quad y_s(1) = A\sin\theta_0 + 0 = A\sin\theta_0$

$y_c(1) = A\cos\theta_0 - \sin\theta_0.0 = A\cos\theta_0$

$y_c(1) = A\cos\theta_0$

n = 2 $\quad y_c(2) = \cos\theta_0 y_c(1) - \sin\theta_0 y_s(1)$

$y_c(2) = A\cos^2\theta_0 - A\sin^2\theta_0$

$y_c(2) = A\cos 2\theta_0$

$\downarrow$

$y_c(n) = A\cos n\theta_0$

(d) $\quad H(z) = \dfrac{2z^3 - 4z^2 + 9}{(z-3)(z^2 + z + 0.5)}$

Since H(z) has a pole at z = 3, the given transfer function is unstable.

To construct a stable transfer function having the same magnitude response consider another transfer function G(z).

$$G(z) = \frac{2z^3 - 4z^2 + 9}{(1-3z)(z^2 + z + 0.5)} = H(z)A(z)$$

where $A(z) = \dfrac{1-3z}{z-3}$ is an all pass function.

Thus $|G(\theta)| = |H(\theta)||A(\theta)|$

$\quad\quad\quad

Question 3:

(a) (i) The monotonic requirements indicate a Butterworth filter.

$$\theta_c = 2\pi \frac{f_c}{f_s} = 2\pi \frac{10{,}000}{100{,}000} = 0.2\pi$$

$$\theta_h = 2\pi \frac{f_h}{f_s} = 2\pi \frac{20{,}000}{100{,}000} = 0.4\pi$$

Prewarping $\omega = \frac{2}{T} \tan \frac{\theta}{2}$

$$\omega_c' = \frac{2}{T} \tan \frac{0.2\pi}{2} = 2f_S \tan(0.1\pi) = 0.6498 \times 10^5$$

$$\omega_h' = 2f_s \tan\left(\frac{0.4\pi}{2}\right) = 1.4531 \times 10^5$$

Butterworth Filter Design

$$|H_a(\omega)|^2 = \frac{1}{1 + \left(\frac{\omega_h'}{\omega_c'}\right)^{2n}}$$

To ensure 10dB attenuation at ω_h'

$$10 \log |H_a(\omega)|^2 = -10 \log \left[1 + \left(\frac{1.4531}{0.6498}\right)^{2n}\right]$$

$$||$$

$$-10 \text{ dB}$$

$$10 = 1 + \left[\frac{1.4531}{0.6498}\right]^{2n}$$

$$\therefore n = 1.367 \qquad \{\text{Select filter order n = 2}\}$$

(ii) Transfer function of the digital Butterworth filter

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \qquad H(s) = \frac{(\omega_c')^2}{s^2 + \sqrt{2}\omega_c' + (\omega_c')^2} \qquad s \to \frac{2}{T} \cdot \frac{1 - z^{-1}}{1 + z^{-1}}$$

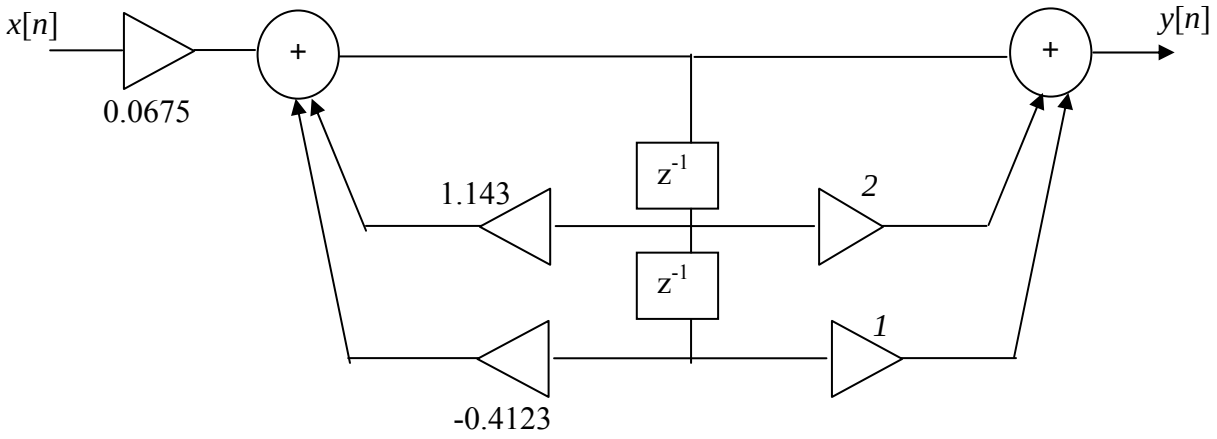
$$H(z) = \frac{\left(\frac{2}{T}\right)^2 (\tan(0.1\pi))^2}{\left(\frac{2}{T}\right)^2\left(\frac{1-z^{-1}}{1+z^{-1}}\right)^2 + \sqrt{2}\left(\frac{2}{T}\right)\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\tan(0.1\pi) + \left(\frac{2}{T}\right)^2 \tan(0.1\pi)^2 + \left(\frac{2}{T}\right)\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$= \frac{[\tan(0.1\pi)]^2 (1+z^{-1})^2}{(1-z^{-1})^2 + \sqrt{2}\tan(0.1\pi)(1-z^{-2}) + [\tan(0.1\pi)]^2 (1+z^{-1})^2}$$

$$= \frac{0.1056(1+2z^{-1}+z^{-2})}{1-2z^{-1}+z^{-2}+0.46-0.46z^{-2}+0.1056+0.211z^{-1}+0.1056z^{-2}}$$

$$= \frac{0.1056(1+2z^{-1}+z^{-2})}{1.5656-1.789z^{-1}+0.6456z^{-2}} = \frac{0.0675\,(1+2z^{-1}+z^{-2})}{1-1.143z^{-1}+0.4123z^{-2}}$$

(iii) Canonic realisation of G(z):



(b) $H(z) = \frac{3 + 3z^{-1}(e^{-2T} - 2e^{-T})}{(1 - e^{-2T}z^{-1})(1 - e^{-T}z^{-1})}$

$$= \frac{A}{1 - e^{-2T}z^{-1}} + \frac{B}{1 - e^{-T}z^{-1}}$$

$$H(z) = \frac{6}{1 - e^{-2T}z^{-1}} - \frac{3}{1 - e^{-T}z^{-1}} \qquad \text{NOTE: } \frac{1}{s+a} \rightarrow \frac{1}{1 - e^{-aT}z^{-1}}$$

$$\therefore H(s) = \frac{6}{s+2} - \frac{3}{s+1} = \frac{3s}{(s+1)(s+2)}$$

Question 4:

(a) (i) N = 7 (given)

$$\phi(\theta) = -\beta\theta = -\left(\frac{n-1}{2}\right)\theta = -\left(\frac{7-1}{2}\right)\theta = -3\theta$$

$$\therefore \beta = 3$$

(ii) Impulse response:

$$h_d(n) = \frac{1}{2\pi}\int_{\frac{-\pi}{7}}^{\frac{\pi}{7}} e^{-j3\theta} e^{jn\theta}\, d\theta$$

$$= \frac{1}{2\pi j(n-3)}\left[e^{j(n-3)\frac{\pi}{7}} - e^{-j(n-3)\frac{\pi}{7}}\right] \qquad n \neq 3$$

$$= \frac{\sin(n-3)\frac{\pi}{7}}{\pi(n-3)} \qquad n \neq 3$$

$$h_d(3) = \frac{1}{7} \text{ for } n=3$$

(iii) Coefficients of the FIR filter using the Bartlett window:

$$h(n) = h_d(n)W_B(n)$$

$$\left.\begin{aligned} W_B(0) &= 0 \\ W_B(1) &= \frac{1}{3} \\ W_B(2) &= \frac{2}{3} \\ W_B(3) &= 1 \\ W_B(4) &= \frac{2}{3} \\ W_B(5) &= \frac{1}{3} \\ W_B(6) &= 0 \end{aligned}\right\} \text{ Bartlett Window}$$

$$h(0) = h_d(0)W_B(0) = 0$$

$$h(1) = h(5)$$

$$h(2) = h(4)$$

$$h(3) = h_d(3)W_B(3) = \frac{1}{7}$$

$$h(4) = h_d(4)W_B(4) = 0.921$$

$$h(5) = 0.041$$

$$h(6) = 0$$

(b) Frequency response of the overall system G(θ) in terms of H(θ)

$$h(n) \xrightarrow{DTFT} H(\theta)$$

$$v(n) = x(-n) * h(n)$$

$$u(n) = v(-n) = x(n) * h(n)$$

$$y(n) = \{h(n) + h(-n)\} * x(n)$$

$$Y(z) = \{H(z) + H(z^{-1})\} \cdot X(z)$$

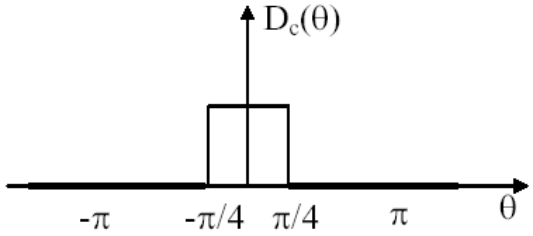
$$G(z) = H(z) + H(z^{-1})$$

$$G(\theta) = H(\theta) + H^{*}(\theta) = (a + jb) + (a - jb) = 2a$$

Hence the equivalent transfer function $G(\theta)$ is real and has zero phase.

Question 5:

(a) (i) $B_c(\theta)=1, D_c(\theta)=1$



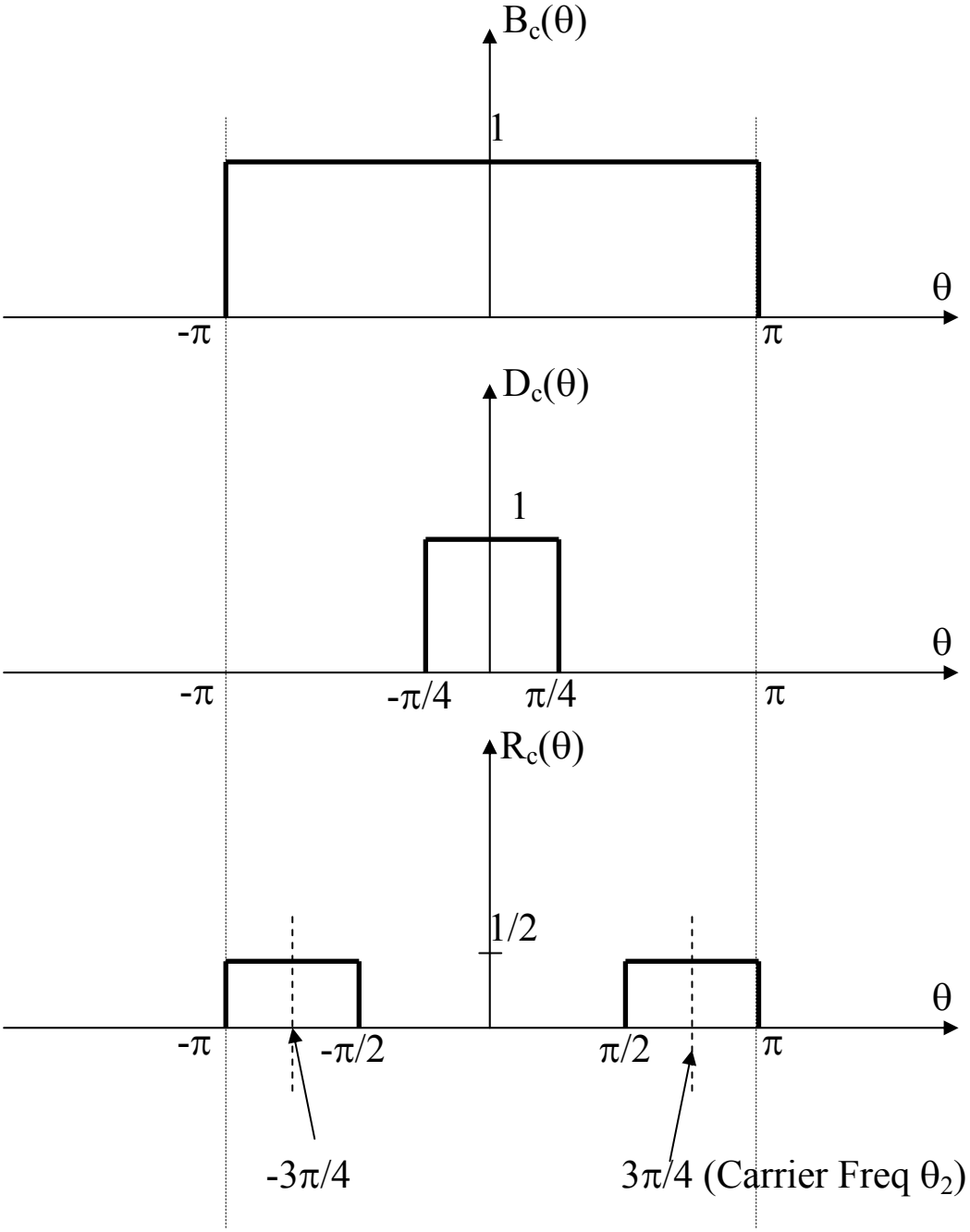
$$R_c(\theta)=\frac{1}{2}\{D_c(\theta-\theta_1)+D_c(\theta+\theta_1)\}$$

$$=\frac{1}{2}\left\{D_C(\theta-\frac{3\pi}{4})+D_c(\theta+\frac{3\pi}{4})\right\}$$

$$=\begin{cases}\frac{1}{2} & \frac{-\pi}{4}(\theta-\frac{3\pi}{4})\le\frac{\pi}{4}\\ \frac{1}{2} & \frac{-\pi}{4}\le\theta+\frac{3\pi}{4}\le\frac{\pi}{4}\\ 0 & \text{elsewhere}\end{cases}$$

$$=\begin{cases}\frac{1}{2} & \frac{\pi}{2}\le\theta\le\pi\\ \frac{1}{2} & -\pi\le\theta\le\frac{-\pi}{2}\\ 0 & \text{elsewhere}\end{cases}$$

(ii) Sketch of $|R_c(\theta)|$ vs θ



(b)

$$P(z) = X(z)z^{-1} \implies P\left(z^{\frac{1}{2}}\right) = X\left(z^{\frac{1}{2}}\right) \cdot z^{-\frac{1}{2}}$$

$$\therefore P\left(-z^{\frac{1}{2}}\right) = X\left(-z^{\frac{1}{2}}\right) \cdot (-1)z^{-\frac{1}{2}}$$

$$W(z) = \frac{1}{2}\left[P\left(z^{\frac{1}{2}}\right) + P\left(-z^{\frac{1}{2}}\right)\right] = \frac{1}{2}\left[X\left(z^{\frac{1}{2}}\right) \cdot z^{-\frac{1}{2}} - X\left(-z^{\frac{1}{2}}\right) \cdot z^{-\frac{1}{2}}\right]$$

$$V(z) = \frac{1}{2}\left[X\left(z^{\frac{1}{2}}\right) + X\left(-z^{\frac{1}{2}}\right)\right]$$

$$Q(z) = V\left(z^2\right) = \frac{1}{2}\left[X(z) + X(-z)\right]$$

$$R(z) = W\left(z^2\right) = \frac{1}{2}\left[X(z) \cdot z^{-1} - X(-z) \cdot z^{-1}\right]$$

$$Y(z) = U(z) + R(z) = z^{-1}Q(z) + R(z)$$

$$Y(z) = \frac{z^{-1}}{2}\left[X(z) + X(-z)\right] + \frac{z^{-1}}{2}\left[X(z) - X(-z)\right] = X(z)z^{-1}$$

$$\therefore y(n) = x(n-1)$$